

RISHIK SARKAR

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EDUCATION

Cornell University <i>Master of Engineering in Computer Science</i>	Aug 2024 – May 2025 GPA: 3.9/4.0
Rutgers University-New Brunswick <i>Bachelor of Science in Computer Science (Honors), Cognitive Science</i>	Sep 2020 – May 2024 GPA: 3.9/4.0

EXPERIENCE

Software Development Engineer <i>Amazon</i>	Aug '25 – Present <i>Boston, MA</i>
- Owned and evolved the Contracts & Agreements platform for multi-country employee contract workflows, processing millions of records/events with strict legal, security, and audit requirements. - Built event-driven distributed systems on AWS using DynamoDB, SQS, and Lambda; shipped APIs/workflows with idempotency, concurrency control, and throttling/backpressure for reliability at scale. - Automated Internal Transfers commit workflow operations by building a script to abort & redrive transfer messages on alarm; reduced incident triage from 20 minutes to <5 seconds via a single time-range/ID input. - Co-led the launch of Collective Bargaining Agreements for Italy, partnering with stakeholders to drive UAT, align HLD across frontend/backend/database, and deliver production rollout under compliance requirements. - In DPX Metadata, owned Amazon-wide builder metadata and helped agentify the builder landing experience via Aza+MCP servers for workflow automation.	
Software Engineer <i>Instalily AI (Google AI Accelerator)</i>	Mar '25 – Aug '25 <i>New York, NY</i>
- Engineered a RAG-based agent and MCP server for United Rentals using Azure AI Search and a custom LangChain-style framework; optimized embedding pipelines and vector index queries, reducing average retrieval latency by 40%. - Built Twilio-powered voice and messaging agents on GCP using Gemini Live API, Azure Search, and ADK/MCP; improved routing and outreach automation, reducing rental conversion time by 50% and driving 40% of sales via autonomous outreach. - Created an autonomous Jira issue resolution agent using GitHub APIs, semantic search, and CI/CD pipelines, enabling fully automated ticket handling in minutes and improving DevOps turnaround time.	
Technical Research Consultant <i>Princeton University (CCNP)</i>	Sep '23 – Jun '25 <i>Princeton, NJ</i>
- Engineered a Python-based ETL pipeline to consolidate 800+ Excel, CSV, and JSON files into a structured SQLite database with enforced schema rules, enabling standardized data for 1000+ subject records. - Built a Tkinter GUI for 15+ researchers to perform SQL queries and CRUD operations, reducing manual data retrieval time and improving access to analytical workflows via Pandas. - Upgraded to a Dockerized Django/PostgreSQL EMR platform with modular workflows and secure web access, enabling faster agile collaboration cycles and improving system scalability across research teams.	
Full-Stack Developer Intern <i>Provenir (Fintech)</i>	Jun '23 – Dec '23 <i>Parsippany, NJ</i>
- Built an automated credit-risk decision engine using Decision Trees, Random Forests, XGBoost, and RNNs via scikit-learn and TensorFlow, achieving 95% accuracy on historical customer data. - Improved model transparency using SHAP/LIME, wrote 100+ unit tests with MockMvc, and containerized deployments on Minikube, boosting system reliability by 20% and reducing latency by 98%.	

TECHNICAL SKILLS

Languages: Python, Java, C#, Kotlin, TypeScript, C/C++, SQL
Frameworks: Spring, Flask, Next.js, Tailwind CSS
Cloud/DevOps: AWS, GCP, Docker, Kubernetes, Jenkins, CI/CD
Databases/Tools: PostgreSQL, MySQL, MongoDB, SQLite, Git, Jira